Drift-Flex v3 — Production Variant Tech Brief

Width-parameterized cryptographic state on SKY130HD · PPA 63/929,897 (Elastic State Extension via Temporal Carry Propagation) · Phase 0 silicon-measurement campaign 2026-05-29

Silicon-measured production variants (SKY130HD, AREA-0, 13.56 MHz)

Variant	Virtual W	GE	GE/bit	Cycles/step	Security tier
drift_flex_v3_p32_v64	64	1,219	19.0	5	Sub-cryptographic (aggregation required)
drift_flex_v3_p32_v128	128	2,064	16.1	9	CNSA 1.0 minimum (smart-dust-adjacent)
drift_flex_v3_p64_v256	256	3,841	15.0	9	CNSA 2.0 IoT minimum
drift_flex_v3_p64_v512	512	6,799	13.3	17	High-classification / vault
drift_flex_v3_p64_v1024	1024	12,667	12.4	33	Deep cryptanalytic margin

Per-virtual-bit cost falls monotonically with W as fixed overhead amortizes (19.0 → 12.4 GE/bit across the 16× width range). W=1024 hits the SKY130-FF architectural floor (bottom-up estimate: ~12 GE/bit). **The W=1024 tier is deep cryptanalytic margin beyond CNSA 2.0 vault — not a quantum hedge.** Symmetric crypto is already PQC-safe at W=256, where Grover halving gives 128-bit-effective security (the standard post-quantum bar for symmetric primitives); Shor’s algorithm does not apply to symmetric crypto. W=512 and W=1024 provide headroom against potential future *classical* cryptanalytic advances against symmetric ciphers themselves. Bit-exact equivalence to a native width-W reference validated via C oracle over 5×10^6 iterations (5 configurations × 10^6 each, all PASS).

Competitive landscape at the cryptographic-minimum tier (128-bit security)

Primitive	Security	Footprint (typical)	PDK / process	Notes
PRESENT-80	80-bit	~1,000 GE	UMC 0.18 µm	Sub-CNSA-1.0
drift_flex_v3_p32_v128	128-bit	2,064 GE	SKY130HD	Measured 2026-05-29
SIMON / SPECK (smallest)	128-bit	~1,500–2,500 GE	Various 130–180 nm	NSA lightweight
Drift Core W=128 (single-cycle)	128-bit	2,828 GE	SKY130HD	Drift agile-state baseline
Ascon (smallest reported)	128-bit	~2,300–3,000 GE	Generic CMOS	NIST LWC final standard
AES-128 (smallest reported)	128-bit	~3,000–3,500 GE	Various	FIPS-validated lineage

Validation and provenance

Bit-exact equivalence: C oracle (`drift_flex_v3_oracle.c`) validates that the chunked computation produces output bit-exactly identical to a native wide-W reference simulator across 5 configurations × 10^6 iterations each. All configurations PASS at 5×10^6 total verifications.

Formal verification: The W-parameterized recurrence has 1 cycle of Lean 4 implementation-safety proofs (determinism, eventual periodicity, recurrent-set cardinality bound) inherited at every supported state width — see `formal_verification/DriftRecurrence.lean + AgileState.lean` in the public drift-core-sim repository. The agile-state mask-gated wide-adder RTL is proved bit-exact with the native modular recurrence at every width (`driftStepAgileRaw_eq`); the W=128 case is the backward-compatibility guarantee with the fixed-width `dad_core`.

Honest competitive framings (literature-validated)

At W=128: We are not aware of a published 128-bit-security symmetric primitive smaller than 2,064 GE on SKY130 specifically. PRESENT-80 (~1,000 GE) provides only 80-bit security on UMC 0.18 µm. SIMON/SPECK at 128-bit security range from ~1,500–2,500 GE on commercial 130–180 nm PDKs; direct GE comparisons across PDKs are approximate due to cell-library normalization differences. The Drift-Flex v3 W=128 variant undercuts the bare Drift Core baseline at the same security tier by 27%.

At W=1024: We are not aware of a published 1024-bit symmetric cryptographic state silicon-measured below 12,667 GE on any open-source standard-cell PDK. Wider-state symmetric primitives (Keccak-f[1600], SHA-512 1024-bit message schedule) exist on commercial nodes; the open-PDK + silicon-measured + 1024-bit-symmetric-state combination is the specific novelty here.